

Assignment Day 3 - Loops

Print 1 to N

```
#include <iostream>
using namespace std;

int main() {
    int n;
    cin >> n;
    for(int i=1;i<=n;i++) cout << i << " ";
    return 0;
}
```

Input: 5

Output: 1 2 3 4 5

Sum of N Numbers

```
#include <iostream>
using namespace std;

int main() {
    int n, sum=0;
    cin>>n;
    for(int i=1;i<=n;i++) sum+=i;
    cout<<sum;
    return 0;
}
```

Input: 5

Output: 15

Factorial

```
#include <iostream>
using namespace std;

int main() {
    int n; long long fact=1;
    cin>>n;
    for(int i=1;i<=n;i++) fact*=i;
    cout<<fact;
    return 0;
}
```

Input: 5

Output: 120

Reverse Number

```
#include <iostream>
using namespace std;

int main() {
    int n, rev=0;
    cin>>n;
    while(n>0){ rev=rev*10+n%10; n/=10; }
    cout<<rev;
    return 0;
}
```

Input: 1234

Output: 4321

Count Digits

```
#include <iostream>
using namespace std;

int main() {
    int n, count=0;
    cin>>n;
    while(n>0){ count++; n/=10; }
    cout<<count;
    return 0;
}
```

Input: 12345

Output: 5

Prime Number Check

```
#include <iostream>
using namespace std;

int main() {
    int n; bool prime=true;
    cin>>n;
    if(n<=1) prime=false;
    for(int i=2; i<=n/2; i++){
        if(n%i==0){ prime=false; break; }
    }
    cout<<(prime?"Prime Number":"Not Prime Number");
    return 0;
}
```

Input: 13

Output: Prime Number